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SCIENTIFIC PRODUCTIVITY AND THE REWARD STRUCTURE OF SCIENCE*

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Regression analyses of longitudinal data for a probability sample of chemists provide estimates of the causal links between chemists' predoctoral training, early productivity, recognition and organizational context and productivity at the end of the first postdoctoral decade. After identifying certain methodological problems involved in using Science Citation Index citation counts to assess the effects of collegial recognition on later productivity, I report tests of several hypotheses proposed to account for scientists' conformity to productivity norms. Apart from the calibre of the Ph.D. department, measures of socialization have no direct effect on decade productivity. Early productivity and collegial recognition do contribute to decade productivity, but the strength of their effects varies by the research orientation of the first employer: early productivity is more important for those employed in universities, whereas collegial recognition is particularly important for chemists in contexts that do not stress scholarly publication.

Scientific research is fraught with difficulties which are not characteristic of most professional work. Scientists must choose research questions that are solvable yet nontrivial, must apply disciplined and creative efforts to intractable problems, must convince others that their results are valid and valuable and must tolerate high levels of competition that may deprive them of the ultimate reward of recognition from the scientific community (Merton, 1957; Hagstrom, 1974). Furthermore, scientists must work without the structural supports for role performance that many other professionals take for granted. For example, scientists' services are not subject to the high public demand that structures the work of free professionals; they must schedule their own work and that of their assistants. Rarely subject to regular supervision, sci-

entists also lack the short-run incentives that rewards for daily performance provide. While they are powerful sanctions in the long run, promotion, tenure and salary raises are probably not especially effective in maintaining day-to-day conformity to productivity norms. Moreover, many scientists have nonresearch obligations (teaching, administration) that do have deadlines, are subject to positive and negative sanctions and, thus, can divert them from their less structured research work.¹

Several social-control mechanisms ensure that some scientists undertake the uncertain task of original research. These include selective recruitment of highly-talented individuals to scientific careers (Harmon, 1965), intensive research training (Ben-David, 1971:141-2), socialization to norms emphasizing research and publication (Hagstrom, 1965:10) and rewards for productivity (Merton, 1957; Hagstrom, 1965). Some observers of science have stressed the intrinsic psychological rewards of scientific achievement, but the former are probably

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¹ Stinchcombe (1966:30) suggested strategies for overcoming these obstacles, but their general effectiveness is still unproved (Hargens, 1977).

not sufficient for most scientists without the professional rewards that usually accompany them. Extrinsic rewards—formal and informal recognition (Hagstrom, 1965); material rewards, such as positions and research resources; and, less frequently, prizes and honors (Zuckerman, 1970:237)—appear to be the major source of social control in science. Yet, in spite of these rewards and their effectiveness for some, many scientists do no original research after leaving graduate school and rarely or never publish (Price, 1963:47). Apparently, the scientific reward-structure, while perhaps sufficient to maintain scientific progress (Cole and Cole, 1972), is not particularly effective in maintaining conformity to professional norms at the individual level. Several hypotheses (reviewed below) have been proposed to account for this variation in scientists' productivity. This paper reports tests of some of them, using data for scientists from a single discipline, chemistry.

DETERMINANTS OF SCIENTIFIC PRODUCTIVITY

Until recently, most studies of scientific productivity were done by psychologists interested in either creativity (Eiduson, 1962) or organizational characteristics that maximize productivity (Taylor and Barron, 1963; Pelz and Andrews, 1966). The former studies, which emphasized the role of personal characteristics such as intelligence, motivation or other personality traits (Roe, 1952), have added little to our understanding of the determinants of productivity. Despite its intuitive appeal, measured ability has yet to distinguish between more and less productive scientists (Taylor and Barron, 1963; Bayer and Folger, 1966:389–9; Cole, 1974:40). Nor have hypotheses about the impact of motivation on productivity been supported (Pelz and Andrews, 1966:190–1). For example, the "sacred-spark" hypothesis (Cole and Cole, 1973:114), which holds that an inner drive compels some scientists to do research even in the absence of rewards, has been untested, largely because of measurement difficul-

ties. Nevertheless, it has been proposed as an alternative explanation for productivity differences unexplained by reinforcement (see, for example, Allison and Stewart, 1974). The Coles (1973:71) have also suggested that sheer stamina may play an important role in scientific productivity, but the relationship between one measure of stamina—number of hours worked per week—and scientific output is weak (Fulton and Trow, 1974:62; Hargens, 1977).²

Socialization

Because some aspects of professional socialization are public, certain facets of socialization are more amenable to measurement than the professional commitment it is thought to engender. Thus, while we lack data that permit exploring the links between socialization, scientific commitment and conformity to professional norms, researchers have studied the relationship between scientists' socialization and their productivity. Aran and Ben-David (1968), for example, found that "resocialization" into the international scientific community was associated with both increased and "more specific" publications among Israeli medical researchers. Reported links between scientists' later success and their sponsors' eminence (Crane, 1965; Zuckerman, 1967) or the calibre of their Ph.D. departments (Bayer and Folger, 1966:388; Hargens and Hagstrom, 1967; Cole, 1974) provide indirect evidence for the effect of socialization, but these associations may also reflect a selection process wherein graduate students' talents are roughly matched with the calibre of their doctoral sponsors or the quality of their Ph.D. departments. Measures of scientists' pre-doctoral ability are necessary to sort out these two processes.

² The empirical support for relationships between other research practices and productivity, while mixed, has been generally negative (Gaston, 1970; Simon, 1974). Variation in four research practices (patterns of collaboration, deadlines, number of projects simultaneously pursued, and number of hours spent on research) accounted for only 5% of the variation in university scientists' output (Hargens, 1977).

Reinforcement and the Accumulation of Advantages

While socialization and professional commitment undoubtedly influence conformity to productivity norms, neither are sufficient to maintain conformity without a reward structure that reinforces appropriate role behavior (Hagstrom, 1965:12, 21). Institutional features of science provide a variety of reinforcers such as the personal satisfaction derived from completing creative work (or, in Guston's [1973] terms, from "partaking of the charisma of science"), collegial recognition expressed through the acceptance of research papers for publication or citations to published work, material rewards such as research grants and the conferral of formal honors (Hagstrom, 1965:28). Given its public nature, formal collegial recognition should be a particularly effective reinforcer. Using citations as a measure of collegial recognition, the Coles (1973:110-5) found some support for the reinforcement hypothesis among university physicists, thereby simultaneously demonstrating the utility of a reinforcement approach and the potential value of more fully specified reinforcement models. Their accumulative-advantage hypothesis (Cole and Cole, 1973:119-22) elaborates the reinforcement hypothesis to take into account the feedback relationship between productivity, recognition and resources. Briefly, collegial recognition of published work demonstrates a scientist's merit to those who distribute professional resources (appointments, promotions, grants, research assistance), which not only reinforce past productivity but also facilitate future productivity. Through this process, successful scientists accumulate rewards that lead to even greater productivity. The "Matthew effect" (Merton, 1968) refers to such disproportionate recognition of the contributions of well-known scientists. Allison and Stewart (1974) hypothesized that the process of accumulative advantage would give rise to an improved fit between resources, productivity and recognition with scientists' increasing age. Their findings for academic scientists were consis-

tent with predictions derived from the reinforcement and accumulative advantage hypotheses. While these studies are highly suggestive, they did not separate the effects of ability and socialization from those of reinforcement. Moreover, they were limited to academic scientists. But reward systems are not uniform throughout the scientific community and their effects may depend on scientists' location. Thus, in order to understand the determinants of conformity to productivity norms, we must consider organizational contexts with varying reward structures.

Organizational Contexts and Rewards

Theoretically, organizational context should directly affect scientists' productivity and specify the effects of other variables. Organizations that value the production of knowledge should reinforce such behavior with professional rewards that will facilitate future research (Hagstrom, 1965:39). Organizations committed to other goals will rely on organizational rather than professional incentives, such as promotion to administrative jobs, to maintain performance (Kornhauser, 1962:131; Hagstrom, 1965:37).

The relationship between setting and productivity is well established (Pelz and Andrews, 1966:6; Fulton and Trow, 1974:46; Hagstrom, 1968) and, as expected, publication is less important for promotion in nonresearch-oriented organizations (Meltzer, 1956). Of course, the association between context and output also might be due to selective recruitment, and this alternative must be considered. In either case, the hypothesized impact of collegial recognition on scientific productivity may vary depending on the values of the employing organization. While it may be attenuated among scientists working in contexts not oriented toward the production of scientific knowledge, scientists in such settings who do publish may be even more responsive to mechanisms that symbolize their connection to the larger scientific community.

THE PRESENT STUDY

The research reported here extends previous work in several ways. First, unlike most studies which used cross-sectional data, this analysis employs longitudinal data for the sample members' first postdoctoral decade, so the time order of important events is clear. Second, previous tests of the reinforcement hypothesis were restricted to that minority of scientists employed in universities (Office of Scientific Personnel, 1967:115).³ While the latter do write most scholarly articles (Krause, cited in Hagstrom, 1965:38), by sampling all scientists who are *qualified to do research* (i.e., Ph.D.s) variation in productivity is increased and the impact of organizational context can be assessed.⁴ Third, data on predoctoral ability and educational background provide controls for some aspects of selection and socialization.

DATA AND MEASURES

This research is based on the careers of a systematic random sample of 238 doctoral chemists who obtained their Ph.D.s from U.S. universities between 1955 and 1961. The sample was drawn from the American Chemical Society's biennial *Directory of Graduate Research (DGR)*, which also provided the names of the chemists' doctoral sponsors. Sample members' educational and career histories were obtained from the 11th and 12th editions of *American Men and Women of Science (AMWS)* and from mailed questionnaires for chemists not included in *AMWS*. These sources ultimately yielded professional histories for 95% of the sample,⁵ including data on length of doctoral

study, and employment setting at the beginning and end of the first postdoctoral decade. In order to examine the impact of context on productivity, I classified organizations in terms of research orientation with two dummy variables: university employment versus all other contexts,⁶ and employment in a research-oriented context (university, government laboratory or nonprofit research institute) versus all other contexts.

Productivity and Recognition Measures

For this study, unpublished work, books and patents were excluded from the productivity measure.⁷ Data on productivity came from *Chemical Abstracts (CA)*, American Chemical Society) and *Science Citation Index (SCI)*, Institute for Scientific Information).⁸ Both sources list articles for all authors of multiple-authored papers. Productivity was observed for three time-periods: prior to the Ph.D. ("predoctoral productivity"), during the three-year period following the Ph.D. beginning with the Ph.D. year ("early productivity") and during the chemists' ninth and tenth postdoctoral years ("decade productivity").

Collegial recognition is usually measured with citation counts (Bayer and Folger, 1966; Cole and Cole, 1971), available in the *SCI* since 1961. *SCI* attributes citations to first authors only. While this procedure apparently does not bias estimates of recognition for established scientists (Hagstrom, 1968; Cole and Cole, 1971), counts for scientists who are disproportionately junior authors will underestimate the total citations to their published work and, presumably, the recognition they experience (assuming, of course, that junior authors feel as rewarded by citations or receive as many rewards for

³ For example, in chemistry, only one-seventh of the Ph.D.s in the late 1950s and early 1960s began their careers in university positions.

⁴ This design also avoids the problem that inter-organizational mobility poses for studies restricted to university scientists.

⁵ Nonrespondents (i.e., those for whom career data were unavailable) and respondents did not differ significantly on either early productivity or their number of early citations, but nonrespondents had significantly fewer decade publications. This association is partly an artifact of the way chemists not included in *AMWS* were located (see Reskin, 1973: Appendix 1). Given the high response rate, this difference should have little effect on the results.

⁶ Academic institutions shown in the *DGR* as granting advanced degrees in chemistry are coded as universities; all others are coded as colleges.

⁷ Books represent a very small part of sample members' publications. Excluding patents, however, does underestimate an important form of the productivity of industrial chemists.

⁸ Article counts came from *CA* through 1964; after 1964, they were taken from *SCI*. I did not use available *SCI* data for 1961 or 1964 because these first two volumes underreport sample members' publications.

them as do senior authors). New Ph.D.s are probably more likely than mature scientists to coauthor papers and more likely to be junior authors.⁹ Thus, citation counts may systematically underestimate the amount of recognition young scientists receive.¹⁰ Ideally, one would identify the first authors of chemists' junior-authored papers and add their citations to the junior authors' totals, but this was not feasible. As an alternative, I took authorship order into account in the analyses reported below. The number of citations to the chemists' early work that were shown in the 1961 or 1964 volumes of *SCI* indicate *early recognition*.¹¹ All productivity and recognition measures were standardized by the chemists' Ph.D. year to adjust for variation in *CA* and *SCI* coverage over time.¹²

Ability and Socialization Measures

Controls for predoctoral ability are desirable both to rule out selection as an alternative explanation for apparent socialization or context effects and to specify the regression equations as fully as possible. Since intelligence-test scores are not related to scientific productivity, they are not a desirable measure of predoctoral ability. But scientists do vary in their propensity to do publishable research; and this trait, which should appear rather early in chemists' training, is fairly stable over time (Hargens et al., 1976). Thus, measures of chemists' predoctoral research output should reflect this propensity. Predoctoral publication has been shown to be related to later productivity among sociologists (Clemente, 1973:415).

⁹ Sample members were junior authors in 44% of their pre-Ph.D. publications and in 35% of the articles published during their Ph.D. year and the two following years. They collaborated with their sponsors on about half of the latter articles.

¹⁰ Previous studies (e.g., Cole and Cole, 1973) have failed to take this problem into account.

¹¹ *SCI* did not publish citation data before 1961 or in 1962 and 1963; thus, I used 1961 citations for 1955 to 1958 Ph.D.s, 1964 citations for 1959 to 1961 Ph.D.s.

¹² Since this procedure precludes interpreting the regression coefficients in terms of the increment in later articles associated with each additional early article or citation, only standardized regression coefficients are reported.

However, simple pre-Ph.D. article counts may be biased in two ways. First, coauthored publications may reflect substantial contributions from more experienced faculty members, so I counted the number of single-authored predoctoral publications. Second, the number of predoctoral publications should reflect the length of time in which predoctoral publication could occur, so that the duration of doctoral study (a second indicator of predoctoral ability discussed below) should be positively related to the number of predoctoral publications. To control for this, I regressed publication counts on the length of doctoral study and used the residuals as a measure of predoctoral ability.

An additional indicator of predoctoral performance is the duration of doctoral study. The negative relationship between time taken to complete the Ph.D. and later professional success observed in several disciplines (Hagstrom, 1971:386; Clemente, 1973:415; Reskin, 1973) has been interpreted as the effect of ability or motivation (Folger et al., 1970:265).

I measured three aspects of socialization. The productivity of the chemists' doctoral sponsors during the period the chemists were graduate students should reflect the socialization that sponsors provide through direct tuition and as role models. Actual student-sponsor collaboration should reflect additional professional guidance beyond that involved in routine supervision. Although faculty differ in their propensity to share responsibility for their students' published research, presumably they have a greater degree of involvement in work which bears their names. Finally, a postdoctoral fellowship may provide additional socialization. Fellowship recipients often study at institutions more prestigious than their Ph.D. departments (Berelson, 1962:127) and with more eminent sponsors (N.A.S., 1969:71), often with the goal of acquiring additional skills (N.A.S., 1969:64-5). And the experience pays off: former fellows are more productive than are nonrecipients (Folger et al., 1970:267; N.A.S., 1974; Reskin, 1976). Sponsors' productivity is the number of articles attributed to the sponsors during the four-year

period ending with the chemists' Ph.D. years minus the number of student-sponsor coauthored papers (the second socialization measure). Postdoctoral fellowships (coded 0 for none, 1 for any) were reported in *AMWS* or the biographical questionnaire.

The calibre of the Ph.D. department (measured by rankings in Cartter, 1966) in part measures the quality of graduate training and is associated with productivity. But this association may also reflect the ability of better-quality departments to recruit the most talented students. Studies that simultaneously considered the impacts of ability and the calibre of the Ph.D. department support interpreting the latter as an indicator of ability (see Bayer and Folger, 1966:388-9; Folger et al., 1970:264; Cole, 1974:70); but without sound measures of ability, the implications of the department-calibre/productivity relationship cannot be resolved.

Finally, productivity varies by specialty within chemistry (with organic chemists publishing less; Blume and Sinclair, 1973:133). Specialty was controlled with a dummy variable with organic chemists coded 0, all others coded 1.

RESULTS

Distribution of Publications

The chemists' distribution of publications is consistent with our knowledge that a small proportion of scientists contribute most publications (Price, 1963:45): 7.5% of the sample members did not publish during their first postdoctoral decade, and an additional 11% published only one article. Only 15% of the sample members authored half of the almost 2,000 articles these scientists published,¹³ and, with one exception, no more than 40% published an article in any year. The Gini index provides a measure of the inequality in the distribution of publications. For the eleven-year period beginning with the Ph.D. year, the Gini indices are remark-

ably stable, about .76, indicating pronounced inequality in article productivity. These values are substantially higher than those Allison and Stewart (1974:600) report, reflecting the greater heterogeneity of my sample.¹⁴

These results lead us to the question addressed in the remainder of this paper: what accounts for the differential productivity of the chemists surveyed? Before reporting analyses for the chemists' productivity at the end of their first professional decade, I present the results of multiple-regression analyses for early postdoctoral productivity and early recognition.

Early Productivity

Regression results in equation (1) of Table 1 show some of the expected effects of predoctoral ability and professional socialization on early postdoctoral productivity. While length of doctoral study did not affect early productivity, the number of single-authored predoctoral publications showed a significant independent effect. The direction of its impact varied with the calibre of the Ph.D. department: for chemists from low-calibre departments (those unranked by Cartter or classified as "not sufficient" or "marginal"), single-authored predoctoral publications were positively related to early postdoctoral output; for chemists from more highly ranked departments, the relationship was negative.¹⁵ This specification by the calibre of the Ph.D. department is reflected by the significant positive main effect of predoctoral publications and the negative effect of the interaction term. Students and faculty collaborated more often in better departments than in low-prestige departments, and those students from high-calibre departments who publish alone may do so for want of faculty coauthors.

¹⁴ The indices for my sample show neither increasing nor decreasing inequality over time. This discrepancy with Allison and Stewart's (1974:600) results is due to the substantially longer period their synthetic cohorts cover.

¹⁵ These findings are based on separate regression analyses for each of Cartter's major department-prestige categories, not presented here.

¹³ Because of my restricted period of observation, this figure is higher than Price's (1963:45-6) estimate of 6-10%; some of the chemists in my sample will never publish after their first postdoctoral decade.

Table 1. Regression Equations for Early Productivity, Early Recognition, and Decade Productivity ^a

Independent Variables	Standardized Regression Coefficients			
	Early Productivity	Early Recognition	Decade Productivity	
	(Equation 1)	(Equation 2)	(Equation 3a)	(Equation 3b)
Prior to Early Productivity				
Single-authored pre-PhD pub.	.251*	.051	.028	.013
PhD duration	.011	-.143*		
PhD calibre	-.028	.112*	.134*	.136*
Sponsors' pub.	-.561*	-.050		
PrePhD collab.	.327*			
Organic specialty	-.164	-.058		
Postdoc. fellow.		.140*	.027	.025
First job univ.	.093		.293*	.288*
PhD dept x spons. pub.	.751*			
PhD dept x prePhD pub.	-.222*			
Prior to Early Recognition				
Early pub.		†	.049	.075
% 1st-authored pub.		†		.120*
1st-authored pub.		.403*	†	†
Pub. x univ. first job		.227*	.446*	.455*
Prior to Decade Productivity				
Early recognition			.330*	.308*
Recog. x univ. first job			-.216*	-.218*
Early pub. x recog.			-.241*	-.245*
Decade job univ.			.291*	.280*
Always univ.			-.228*	-.215*
R ²	.281	.367	.386	.399

^a Except for a single indicator of each theoretically relevant control variable, variables that were not statistically significant at the .05 level were excluded.

* Coefficient at least twice its standard error.

† Omitted in favor of an alternative measure. See text for bases for selecting alternative measures for particular equations.

Two of the socialization indicators significantly affected early productivity, but the effect of one—sponsors' productivity—also depended on the calibre of the sample members' Ph.D. departments. In low-calibre departments, working with a productive sponsor *reduced* the chemists' own early productivity, whereas in "distinguished" departments, having a prolific sponsor increased early postdoctoral output (see footnote 15). This interaction is reflected in the significant negative main effect of sponsors' productivity and the significant positive product of the former and Ph.D.-department calibre shown in equation (1). Assuming a moderately good fit between faculty talents and departmental calibre, faculty at top departments would be better researchers and have more to offer their students than equally productive staff at

low-prestige departments. Productive faculty at lower-quality institutions might also be skeptical about the value of investing time in training students whom they do not expect to pursue research careers.

As predicted, the chemists benefited from predoctoral collaboration with their sponsors. While this result may reflect socialization, an alternative explanation is possible. Collaborations among chemists during graduate school may lead to postdoctoral coauthored articles. To test this alternative, I included the number of articles the chemists coauthored with their sponsors during the first two post-Ph.D. years in the regression equation. While the standardized coefficient for predoctoral collaboration dropped slightly (from .33 to .29), it remained an important predictor of early productivity.

Any socialization resulting from a post-

doctoral fellowship should not be reflected in early productivity because the interval between completing the postdoctoral fellowship and the time remaining in the early-productivity measure was too short for any effect of an award to be reflected.¹⁶

Apart from the interactions described above, the calibre of the Ph.D. department was unrelated to early productivity. (The negative sign for its regression coefficient results from including the two Ph.D.-calibre interaction terms in the equation and has no substantive meaning.) More surprising was the failure of either of the measures of first employment setting to show the expected positive effects.¹⁷ The zero-order associations between setting and productivity, although small ($r = .11$ for universities; $.16$ for all research-oriented organizations), were significant, but their partial association disappears because the same background variables that affect early productivity are important determinants of working in these contexts.

Early Recognition

The analysis of early recognition provides a partial test of the accumulative-advantage hypothesis. Scientists may accumulate advantages by acquiring training or credentials that enhance either the quality or the visibility of their work. The advantages may elicit greater collegial recognition (in itself reinforcing) and may also attract additional resources that will facilitate future performance. For example, a purported benefit of the "Matthew effect" (Merton, 1968) for young scientists is the increased visibility resulting from collaborating with established scientists. Even without actually collaborating, students of well-known scientists may receive disproportionate recognition.¹⁸

¹⁶ In addition, for 29% of the postdoctoral fellows the award was subsequent to the period covered by the early-productivity measure.

¹⁷ The results for employment in a research-oriented organization are similar to those shown in equation (1) for university employment.

¹⁸ This assumes that this ascribed status is known to potential citers; acknowledgements sometimes reveal such information. Of course, such a pattern would also occur if they did better work.

Young chemists who come from prestigious Ph.D. departments or who obtain desirable university jobs may also accumulate advantages if their institutional affiliation listed on their articles elicits increased citations. I tested for such advantages by examining the effects on early citations of the interaction of early productivity and (a) collaborating with one's sponsor, (b) collaborating with a particularly productive sponsor, (c) coming from a prestigious Ph.D. department and (d) working in a university.

The regression equation for early citations is shown in equation (2). Although predoctoral publication was not related to early recognition, the more quickly the chemists completed their training, the more often they were cited early in their careers. The calibre of their Ph.D. department and a postdoctoral fellowship were also positively related to early citations. Their sponsors' productivity did not affect the frequency with which the chemists were cited, and the significant zero-order effect of coauthoring papers with their sponsors during graduate school ($r = .18$) was mediated entirely by its impact on early postdoctoral productivity. However, before concluding that collaboration had no direct effect on recognition, we must ask whether order of authorship specified the relationship. Given the underestimation of citations to junior authors, the expected collaboration effect should hold only for chemists who were primarily senior authors. In fact, results of an analysis of covariance not shown here revealed no significant difference between chemists who were exclusively junior and exclusively senior authors; in neither of these subgroups did collaboration contribute significantly to recognition.

Since cited scientists first-author at least one article (except for a few citations to unpublished work), the strong effect of the number of first-authored articles published between the Ph.D. year and the second postdoctoral year ($b^* = .403$) is expected.

Although employment context had no simple effect on citations, the return in citations to early publications was significantly greater for chemists whose first

jobs were in universities. Their work may have been more useful than that produced by nonuniversity chemists, or their institutional affiliation may have influenced readers' judgments. None of the other product terms included to test accumulative advantage was statistically significant, and they are not included in equation (2). While the significant effect for the employment-context/early-productivity interaction term is consistent with the accumulation of advantages to scientists who were initially well placed, it would also follow from a selection process in which potentially better researchers who produced work deserving greater recognition were hired by universities. Lacking an independent measure of the quality of publications (citations themselves are usually used for this purpose), I can not choose between these interpretations. Still, several measures of ability were controlled in the regression procedure. At the very least, these results do not rule out the possibility of accumulative advantage with respect to the first-job setting.

Decade Productivity

By the end of their first postdoctoral decade, sample members had had ample time to establish research programs, publish results and build at least minor reputations. Thus, I observed their productivity at this point, seven years after early productivity was observed and three to six years after early citations were counted. The lag is small enough for any effects of early recognition to persist, but enough time elapsed so that any independent effect of early productivity would constitute support for the sacred-spark hypothesis. Mobility during the first postdoctoral decade should lead to an improved fit between the chemists' productivity and employers' requirements, so organizational context at the end of the decade should mediate the effects of early productivity on later productivity. The decade-productivity regression results (see equations [3a] and [3b]) provide some support for the sacred-spark, reinforcement, and accumulative-advantage hypotheses.

Predoxal performance and socialization. The calibre of the Ph.D. department

exhibited a significant independent effect on the chemists' productivity at the end of their first professional decade. Its impact is noteworthy, particularly because this variable did not significantly affect early productivity. The calibre of the Ph.D. department might reflect the enduring, albeit slight, impact of access of effective socialization on subsequent productivity. Institutions differ in the quality of instruction they provide, and these differences presumably affect their graduates' later performance. But department calibre may also have ascriptive effects on productivity (Crane, 1965; Hargens and Hagstrom, 1967) through its impact on the allocation of certain unmeasured resources to young scientists. For example, established scientists more often may invite young colleagues from prestigious Ph.D. departments to collaborate, either out of paternalistic feelings or from a desire to exploit the latter's familiarity with the newest techniques.

The importance of these "school" effects is highlighted by the absence of any independent effects of sponsorship. Collaborating with one's sponsor during graduate school led to increased early postdoctoral productivity, but any direct effect was lost by the end of the decade. Similarly, studying under a highly prolific sponsor increased recognition for early work, but had no direct effect on later productivity. This does not mean that the quality of sponsorship is unimportant for scientists' careers, but any measurable effects on productivity may be short-run and primarily ascriptive.

I argued that by providing additional socialization a postdoctoral fellowship should increase conformity to productivity norms and, hence, output; but, contrary to previous findings, the analyses revealed neither short- nor long-run effects. Apparently the effects of the precursors and consequences of a fellowship account for most of its zero-order association ($r = .17$) with decade productivity.

Organizational context. Beginning one's career in a university was a far more important determinant of decade productivity ($b^* = .29$) than one's graduate school credentials, and this effect was independent of university employment at

the end of the decade. Initial university employment could increase the decade productivity of scientists who subsequently left the university for several reasons, the most obvious being the continuation of collaborative work with former university colleagues. Former colleagues may also control the allocation of ongoing opportunities as conference organizers or journal editors, for example, or they may continue to serve unwittingly as a reference group. Some scientists who leave the university for less prestigious jobs may try to show former colleagues that they could have succeeded by university standards. Exposure to professional values regarding the primacy of research on one's first job may also facilitate their internalization (see, e.g., Cotgrove and Box, 1970: 127); chemists whose first jobs are in nonresearch-oriented contexts may view such values as purely academic. Finally, the enduring effect of early university employment may actually reflect universities' ability to recruit (if not always to retain) potentially productive chemists. While graduate students prefer university careers (Davis, 1962:158), in the face of growing economic pressures some of these more able scientists may be seduced by the financial prospects of industry. Controlling for predoctoral ability reduces the plausibility of this alternative, but does not exclude it.

Early productivity and the sacred-spark hypothesis. The data do not permit a direct test of the sacred-spark hypothesis. However, such a drive to publish should be fairly stable, leading to a strong partial association between early and later productivity. We see in equation (3a) that early productivity did not significantly affect decade output, but the absence of a direct effect does not necessarily disconfirm the hypothesis. While the failure to publish early in one's career may signal the absence of a spark, publication does not necessarily signal its presence, because a chemist's early publication rate may reflect the efforts of more experienced or motivated collaborators primarily responsible for joint efforts. If we assume that the order of authorship on articles reflects the authors' relative contributions, the partial association between

early first-authored articles and decade productivity provides a more realistic test of the sacred-spark hypothesis.

When both early productivity and the proportion of early first-authored publications were included in the regression, the latter did positively affect decade productivity (see equation [3b]). Apparently, early-publication rate, when measured to exclude publications for which the chemists may not have been primarily responsible, is associated with subsequent productivity. These results provide *very limited* support for the sacred-spark hypothesis: while their slopes are statistically significant, less than one percent of the variance in decade productivity among the sample as a whole can be attributed to the operation of a sacred spark.

While the sacred-spark hypothesis implies that strongly committed scientists will publish without social support, it does not necessarily follow that social mechanisms and social context have no effect on the productivity of individuals either blessed with or lacking a spark. Among chemists whose first jobs were in universities, early productivity significantly affected decade output. Since the emphasis on and rewards for productivity are generally greater in universities than in other sectors, a drive to publish should flourish in such settings. The significant-setting/early-productivity interaction is also consistent with the accumulative-advantage hypothesis. Universities allocate a variety of unmeasured rewards for early productivity (desirable teaching assignments, increased research facilities, etc.) that presumably foster continued productivity (Hagstrom, 1968). For nonuniversity chemists, other signs of successful performance (teaching evaluations, development of industrial applications, patents) bring similar organizational rewards, while publication is presumably incidental to career advancement.

Collegial recognition and the reinforcement hypothesis. Collegial recognition in the form of citations has a strong impact on decade productivity ($b^* = .308$). Since citation counts underestimate the number of citations for junior authors, I tried to assess the effect of this bias by

examining the relationships among the 98 chemists who had no junior-authored early publications. Among this subsample, the independent effect of early citations was 40% stronger than it was for the entire sample ($b^* = .456$ and $.308$, respectively; the equation for the subsample is not shown). Thus, of the variables considered here, early recognition is the most important determinant of decade productivity, accounting for almost 10% of its variance. However, this probably overestimates the effect of collegial recognition, which is certainly correlated with other, unmeasured structural rewards for early productivity that also facilitate productivity.¹⁹

Hagstrom's (1965:37) discussion of organizational incentives implies that the effect of recognition should vary by organizational context, with recognition from the scientific community being less important to scientists in contexts that do not emphasize research. To test this, I introduced into the regression equation a multiplicative term for first-job context and early citations. While significant, the effect of recognition on later productivity was *weaker* for chemists whose first jobs were in universities or other research-oriented organizations. Apparently in organizations that emphasize and immediately reward research and publication, formal recognition from the scientific community is less important in maintaining conformity to productivity norms than in nonresearch-oriented contexts in which rewards for productivity may be uncertain or scarce.²⁰

The impact of recognition on chemists' later productivity also varies with their

early productivity level. The greater one's early productivity, the weaker the effect of recognition on later output. Citations may be more reinforcing to late bloomers than to chemists whose early publication records are strong. The latter may have a strong drive to publish and need less social support to maintain their output or, perhaps, the intrinsic rewards of their early successes render productive researchers less vulnerable to the reactions of the scientific community.

Accumulative advantage. Without data on the chemists' research resources, I cannot directly test the accumulative-advantage hypothesis, which predicts both a direct reinforcement effect of citations and indirect effects from early productivity and early recognition mediated by the resources both can attract. Since research resources vary considerably by organizational context, a crude test of the hypothesis might use decade context as a proxy for resources. University employment at the end of the decade significantly increased decade productivity ($b^* = .280$), which is consistent with the accumulation of advantages. However, the selection hypothesis remains a viable alternative.

Chemists who began and ended their first decade in universities were significantly less productive than those who moved into universities from other contexts. While this result is consistent with a beneficial effect of mobility, it is as likely that universities require faculty they recruit from other contexts to be superior to their current faculty (see Cole and Cole, 1973:118).

SUMMARY AND CONCLUSIONS

Regression analyses of longitudinal data have provided estimates of the causal links between chemists' predoctoral training, early productivity, collegial recognition, organizational context, and productivity at the end of the first postdoctoral decade. After identifying methodological problems involved in using *SCI* citation counts to assess the effect of collegial recognition, I report tests of several hypotheses that have been proposed to account for variation in scientists' productivity.

¹⁹ The coefficient for citations will better estimate their true effects (in similarly specified equations) in fields where research work is less routine, since in routine fields such as chemistry (Hargens, 1975) the unmeasured resources that should accompany recognition are more likely to lead to increased productivity.

²⁰ A larger proportion of chemistry Ph.D.s work in nonacademic settings than in any other natural science, so nonacademic chemists may feel less isolated from colleagues doing basic research than similarly situated scientists in other disciplines. Insofar as this is true, the impact of formal recognition on nonacademic scientists' productivity should be even stronger in fields in which most practitioners are concentrated in university research.

Apart from the enduring effect of the calibre of the Ph.D. department, the effects of predoctoral socialization are slight and short run. Sponsor's productivity affects only early recognition; and predoctoral collaboration with one's sponsor increases early productivity, but does not affect decade productivity directly. Although sponsorship may ascribe to new Ph.D.s a professional status that affects their job placement and, indirectly, their productivity, educators apparently overestimate the direct effects of graduate training on scientific achievement.

The hypothesis that collegial recognition in the form of citations influences later productivity was supported, but to a smaller extent than past research on university scientists would suggest. This is not due to the heterogeneity of the sample: the effect of citations on decade productivity was *weaker* for those in contexts that emphasize research. In such contexts, immediate, informal recognition from research-oriented colleagues may be more important in maintaining productivity than the formal, but delayed recognition that citations provide. This is consistent with the greater direct effect of early productivity on decade productivity among university chemists than those employed in other contexts. Given the reward structure in most university departments, the act of publishing—signalling both successful professional performance and a variety of forthcoming rewards—may be especially reinforcing to university scientists. On the other hand, scientists in contexts that do not stress research and who thus lack informal or material rewards for publishing, may depend more on the formal response of the scientific community. Such recognition may symbolize their ties to the larger scientific enterprise and thereby encourage continued conformity to productivity norms.

The effect of organizational context points to the role of organization-specific reward structures and is consistent with the accumulation of advantages among scientists whose jobs provide access to resources that facilitate productivity. However, if context only affected performance in this way, its impact should dis-

appear when scientists change employment settings. The enduring effect of beginning one's career in a university may reflect on-the-first-job continuation of predoctoral socialization.

Although the analyses provide no direct evidence for the sacred-spark hypothesis, neither do they permit its disconfirmation. Chemists' predoctoral and early postdoctoral productivity had very small but significant independent effects on decade productivity. Moreover, the greater one's early productivity, the less the importance of recognition for later productivity; thus, highly motivated scientists—or those who quickly developed effective work habits—need less recognition to continue publishing. The expression "sacred spark" may be an unfortunate choice, since it suggests an innate drive uninfluenced by social factors. Any relatively stable propensity to publish probably depends on scientists' achievement needs, socialization and a constellation of competing interests and activities. The effect of such a latent propensity on actual performance would depend on a scientist's circumstances, including access to resources and rewards. The interaction effects on decade productivity of early productivity with both employment context and recognition should remind us that individuals' scientific drives operate in social contexts.

These results bear on the general question of maintaining normative conformity. Theories of normative conformity stress the importance of sanctions to maintain commitment induced by socialization (Goode, 1960: 254). While my results confirm the importance of *positive* reinforcement among professionals, they provide an incomplete picture of the role of sanctions in maintaining role performance. Anticipated or actual *negative* sanctions are also powerful motivators for scientists. Theories of scientific competition, for example, emphasize the consequences of fear of anticipation on research and publication (Collins, 1968; Gaston, 1971; Hagstrom, 1974). Thus, the extent and severity of competition in a scientific specialty may increase scientific productivity; and those factors that affect competition, such as specialty size, normative

consensus and functional integration (Hagstrom, 1974; Hargens, 1975), should, in turn, affect productivity indirectly. A single control for specialty, as used here, cannot capture such differences. Similarly, the publish-or-perish injunction expresses an important potential sanction for academic scientists. The actual motivating capacity of job insecurity should vary with conditions in the job market (which also vary by specialty). Finally, scientist may fear that past or present associates will view them as failures if their productivity declines (Fulton and Trow, 1974:54; Zuckerman, 1967: 402). If scientists publish in part to avoid losing face with significant others, poorly integrated scientists would have less to lose from low performance and, consequently, should exhibit lower productivity (see Homans, 1974, regarding the relationship between social status and conformity). Studies of scientists' actual and perceived positions in collegial networks are necessary to test these conjectures. I have shown that the same rewards vary in their importance with scientists' employment locations and exposure to formal reward structures. Surely integration into informal networks or invisible colleges is as important.

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